

Grain Zero Deepening Pass: Typed Residual Formalization, Threshold Interface, Pair-Overlap Channel, Fan Calibration

Deepens ARX-001 (PEAICE-KAKEYALOGIC-DDATL-002-WHITEPAPER) and ARX-002 (PEAICE-CLAUDEV6-KNS-LB-PAPER-001)

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Session export 2026-07-08 · β -Protocol · II.1 claim discipline · $h < 1$ · RH OPEN ·
Coleman OPEN · $\zeta(0)$ firewall active

0. Session scope and downstream state

This pass executes the directive “See downstream. Deepen and further formalize” against the arXiv-lane scaffold pack (ARX-001, ARX-002, PEAICE-ARXIV-LANE-001). Downstream state was retrieved live: the DDATL-002 ladder transfer (Rungs 1–8), the canon register docs/ddatl-002-grain-zero.md, both manuscript STATUS files with their “Next edits” blocks, and the GZ-D receipts from the 2026-07-06 GWZ integration session. Source verification of Guth–Wang–Zahl arXiv:2601.14411 (v1, 2026-01-20, math.CA) was performed against the public abstract and opening pages.

The pass delivers: (i) an exact excess-mass identity; (ii) a collapse lemma showing Definition DDATL-002.1 is vacuous without a carrier-typing clause, plus the repaired definition and the weak/strong split of Target 002.A demanded by the ARX-001 STATUS file; (iii) a threshold interface compressing ladder Lemmas 2–4 into one coverage statement; (iv) a pair-overlap channel with a source contact point; (v) the excess-spectrum dichotomy with a $d = 2$ theorem and a $d = 3$ open fork; (vi) full proofs of ARX-002 Lemma 3.1 and explicit two-sided witnesses for Proposition 4.1, ready for LaTeX port; (vii) a deepened continuation criterion for $Z_{G_0}(s)$ at Rung 8, firewall intact; (viii) a carrier-pinning registry v0.2 with source-verified rows; (ix) GZ-D wiring; (x) exact pre-submit replacement text. NUMERICS receipts from a deterministic $d = 2$ probe are appended.

Every load-bearing claim below carries a tag and a falsifier. All new FORMAL items are PROPOSED-FOR-CANON pending Grok TERMINAL cross-derivation (evaluator non-sovereignty, $h < 1$). Nothing here is a proof of the Riemann hypothesis or the Coleman conjecture; both remain OPEN.

1. Symbol hygiene and two new symbol-wall rows

Source verification of GWZ Theorem 1.1 confirms the walled loss exponent β_{GWZ} at its address, and surfaces two further source symbols that must be walled off from program usage before ARX submission:

β_{GWZ}	GWZ Thm 1.1 loss exponent	(registered wall, confirmed at source)
η_{GWZ}	GWZ Thm 1.1 tolerance exponent	(NEW wall row – “for every $\beta > 0$ there is $\eta > 0$ ”)
λ_{GWZ}	GWZ shading density $\lambda(T, Y) = \sum Y(T) / \sum T $	(NEW wall row)
Δ_{max}	GWZ / Wolff clustering functional	(import, keep source name)
ζ_{GWZ}	GWZ Lemma 9.1 density-surplus exponent	(registered wall, per 2026-07-06 receipts)

Program consequences. The multiplicity threshold introduced in §4 is written τ (never λ , which is now reserved to the source shading density). The tolerance exponents of the program’s weak form are written as generic $o(1)$ losses, never η . Harmonization: ARX-001 writes multiplicity m_δ while ARX-002 imports μ ; adopt m_δ throughout and note $\mu \equiv m_\delta$ once in ARX-002. Ladder numbering: adopt the six-lemma transfer

form (Lemmas 1–6); the canon register’s five-lemma block folds Carrier Exhaustion into Lemma 2 — one-line edit OWED in docs/ddatl-002-grain-zero.md.

Fixed notation for this pass: T_δ a finite family of δ -tubes in $\mathbb{R}^3$ ($\mathbb{R}^2$ where stated), shading $Y(T) \subset T$, union U , multiplicity $m_\delta(x)$, excess $e_\delta(x) = (m_\delta(x) - 1)_+$, total shading mass $\Sigma Y := \sum_T |Y(T)|$, raw excess mass $E_\delta := \int e_\delta dx$, compression ratio $u_\delta := |U| / \Sigma Y \in (0, 1]$.

2. The excess-mass identity

Identity I.1 (Excess mass identity) — FORMAL. For any finite tube family with measurable shadings,

$$|E_\delta = \int (m_\delta - 1)_+ dx = \sum_T |Y(T)| - |U(T_\delta, Y)|.$$

Proof. $m_\delta = \sum_T 1_{Y(T)}$ pointwise, so $\int m_\delta = \Sigma Y$. Since m_δ takes integer values, $\min(m_\delta, 1) = 1_U$ and $(m_\delta - 1)_+ = m_\delta - \min(m_\delta, 1)$. Integrate. ■

Falsifier: a single configuration violating the displayed equality (none exists; this is an identity, and the appended probe reproduces it to machine precision on a grid).

Corollary I.2 (Compression form) — FORMAL. $E_\delta / \Sigma Y = 1 - u_\delta$. The normalized raw excess is exactly one minus the compression ratio. Consequences: (a) raw excess never vanishes for compressed families — if $u_\delta \rightarrow 0$ (Besicovitch-type compression, theakeya-critical regime) then $E_\delta / \Sigma Y \rightarrow 1$; (b) Target 002.A is trivially true when $u_\delta \rightarrow 1$ (quasi-disjoint shading), since $G_0 \leq E_\delta$ by monotonicity. All content of DDATL-002 lives in the compressed and intermediate regimes, and Rung 6 is unreachable by any argument that shrinks E_δ itself. Carrier subtraction is not a convenience; it is forced. Probe receipt: for the $d = 2$ fan, $E_\delta / \Sigma Y \rightarrow 0.87$, non-vanishing, Appendix A.

3. The typing gap: collapse lemma and the repaired definition

Lemma C.0 (Collapse of the untyped target) — FORMAL. Definition DDATL-002.1 constrains $C_{\{\delta, \rho\}}$ only to be a measurable set produced at some proof stage, with an open-ended structure list. Under that reading, the choice $C = \{x : m_\delta(x) \geq 2\}$ is Borel, stage-agnostic, and gives $dG_0 \equiv 0$ identically, since e_δ is supported on $\{m_\delta \geq 2\}$. Hence Target 002.A, as currently written, is vacuously true for every tube family and every δ .

Proof. $\text{supp } e_\delta = \{m_\delta \geq 2\}$ up to null sets; multiplying e_δ by the indicator of its own support’s complement yields the zero measure. ■

Reading. This is not a defect of the residual idea; it is the precise localization of where the mathematics lives. One hundred percent of the content of Grain Zero is in the carrier-typing constraint. The definition must say which sets are legal carriers, at what cardinality, and by what extraction relation — otherwise the superlevel set of multiplicity absorbs everything for free. Falsifier for C.0: exhibit a stage semantics under which $\{m_\delta \geq 2\}$ is not extractable; none is available in the current text of ARX-001 §3.

Definition 002.1’ (Admissible carrier system) — FORMAL (definition); catalogue pinning OWED. Fix a proof stage σ of the WZ/GWZ reduction and a scale passage $\delta \leq \rho \leq 1$. An admissible carrier system at stage σ is a set $C = \bigcup_{i \leq N} K_i$ satisfying:

K1 (cell type)	each K_i is a convex cell from the stage catalogue
Cat $_\sigma(\delta, \rho)$	(grains, slabs $\delta \times 1 \times 1$, planks, prisms, Katz–Tao convex cells,
	sticky packets, scale-induction envelopes — pinned per §10)
K2 (budget)	$N \leq N_\sigma(\delta, \rho)$, the cardinality budget the stage’s
estimates meter	

K3 (extraction) cluster K _i	each K _i is proof-generated: it carries an assigned tube T _i ⊂ T _δ standing in the stage's factoring relation to
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Write $\text{Adm}_\sigma(\delta, \rho)$ for the family of admissible systems, and define the typed residual as the extremal quantity

$$G_\sigma^{\{\sigma\}}_{\{\delta, \rho\}} := \inf \left\{ \int_{\mathbb{R}^3 \setminus C} e_\delta \, dx : C \in \text{Adm}_\sigma(\delta, \rho) \right\}.$$

The infimum formulation removes the ambiguity of “the” carrier, makes G_σ canonical per stage, and is monotone: enlarging the catalogue or the budget can only lower G_σ . The untyped limit (all measurable sets, no budget) recovers Lemma C.0: $\inf = 0$. K2 is what blocks the degenerate cover — without a budget, unit-scale convex cells tile any superlevel set.

Target 002.A' (typed split) — executes ARX-001 STATUS Next-edit 2.

002.A-weak (PROPOSED) $G_\sigma^{\{\sigma\}}_{\{\delta, \rho\}} / \Sigma Y \leq \delta^{\{o(1)\}} \cdot \max(u_\delta \cdot \tau(\delta), D(\delta)/\Sigma Y)$ $= o(\Sigma Y)$ [§4]	for some $\tau(\delta) = \delta^{\{-o(1)\}}$ and coverage defect $D(\delta)$
002.A-strong (OPEN) uniform in (T_δ, Y)	$G_\sigma^{\{\sigma\}}_{\{\delta, \rho\}} / \Sigma Y \rightarrow 0$ as $\delta \rightarrow 0$, post-GWZ catalogue,

The GZ-D1 retention calculus (every kept refinement holds a $\geq$ polylog fraction) is exactly what licenses the multiplicative $\delta^{\{o(1)\}}$ tolerance in the weak form: losses along the refinement chain compose polylogarithmically, never polynomially. Falsifier for 002.A-weak: a family and stage where every admissible system within budget misses a fixed positive fraction of E_δ at all small δ .

4. Threshold interface: Lemmas 2–4 compress to one coverage statement

Lemma T.1 (Graded threshold bound) — FORMAL. Let $\tau \geq 2$ and let $C \in \text{Adm}_\sigma(\delta, \rho)$. Write $D_\tau(C) := \int_{\{m_\delta \geq \tau\} \setminus C} e_\delta \, dx$ for the coverage defect of C above level τ . Then

$$\int_{\mathbb{R}^3 \setminus C} e_\delta \, dx \leq (\tau - 2) \cdot |U \cap \{m_\delta \geq 2\}| + D_\tau(C) \leq (\tau - 2) |U| + D_\tau(C).$$

Proof. Split the residual over $\{2 \leq m_\delta < \tau\}$ and $\{m_\delta \geq \tau\} \setminus C$. On the first set $e_\delta = m_\delta - 1 \leq \tau - 2$ pointwise; its measure is at most $|U \cap \{m_\delta \geq 2\}|$. The second integral is $D_\tau(C)$ by definition. ■

Corollary T.2 (Reduction of ladder Lemmas 2–4) — FORMAL. Define the superlevel coverage statement

$$SC(\tau): \text{ there exists } C \in \text{Adm}_\sigma(\delta, \rho) \text{ with } D_\tau(C) = o(\Sigma Y) \text{ as } \delta \rightarrow 0.$$

If $SC(\tau(\delta))$ holds for some $\tau(\delta)$ with $\tau(\delta) \cdot u_\delta \rightarrow 0$, then 002.A-strong holds; if it holds with $\tau(\delta) = \delta^{\{-o(1)\}}$ and defect $\delta^{\{o(1)\}} \Sigma Y$, then 002.A-weak holds. Ladder Lemmas 2 (carrier exhaustion), 3 (residual domination) and 4 (residual vanishing) therefore compress into the single two-parameter interface (τ_σ, D_τ) : what a stage must supply is a threshold its estimates can see and a budgeted cover of the superlevel set above it. This is the load-bearing rewrite: the geometric burden is a structure theorem for superlevel sets of m_δ — morally the engine of WZ/GWZ, which converts high multiplicity into convex structure — restated as a mass-accounting obligation.

Lemma B.1 (Carrier volume budget, necessary direction) — FORMAL. If C captures a $(1 - q)$ -fraction of E_δ , then since $e_\delta \leq m_\delta \leq \#T_\delta \approx \delta^{\{-(d-1)\}}$ pointwise,

$$|C| \geq (1 - q) E_\delta / \|m_\delta\|_\infty \geq c (1 - q) E_\delta \cdot \delta^{\{d-1\}},$$

and via Hölder, $|C| \geq ((1 - q)E_\delta)^{\{p\}} \|m_\delta\|_{p^{\{-p\}}}$ for any imported multiplicity estimate $\|m_\delta\|_p$.

Instantiation slots: $p = 2$ with Córdoba ($d = 2$, KNOWN); $p = 5/2$ with Wolff 1995 ($d = 3$, KNOWN); $p = 3$ (the endpoint maximal strengthening in $d = 3$) — whether the GWZ set-theorem yields the L^3 maximal inequality

is NOT asserted here; verification OWED. Budgets in K2 must be compatible with these volume floors; this is the quantitative teeth behind “legal carriers are thin.”

5. Pair-overlap channel: a literature-anchored domination route

Lemma P.1 (Pair-overlap domination) — FORMAL. For integer $m \geq 0$ one has $(m - 1)_+ \leq m(m - 1)$. Hence

$$\left| \begin{array}{l} E_{\delta} \leq \int m_{\delta} (m_{\delta} - 1) dx = \sum_{\{T \neq T'\}} |Y(T) \cap Y(T')| \quad (\text{ordered pairs}). \end{array} \right.$$

Proof. The pointwise inequality is immediate case-by-case ($m = 0, 1$ give $0 \leq 0$; $m \geq 2$ gives $m - 1 \leq m(m - 1)$). The identity expands $m_{\delta}^2 = \sum_{\{T, T'\}} 1_{\{Y(T) \cap Y(T')\}}$ and subtracts the diagonal. ■

Contact point. The right-hand side is the second factorial moment of multiplicity — exactly the pair-incidence mass that the GWZ-adjacent literature manipulates directly: the $\text{Tri}(S)$ measure of the Cohen rigorous write-up (triples (x, S_1, S_2) with $x \in Y(S_1) \cap Y(S_2)$, for $\delta \times 1 \times 1$ slabs) and the product-organized overlap bound recorded at GWZ Prop 5.1(5) / L5.11 in the 2026-07-06 integration receipts. Grain Zero’s raw excess is thus dominated by an object the source proof already estimates. This gives ladder Lemma 3 its first literature-anchored channel; it is coarse (it overweights high multiplicity quadratically) but exact.

Lemma P.2 (Pairwise intersections are convex-carried) — FORMAL. Let T_1, T_2 be δ -tubes about unit segments with direction angle $\theta = \angle(\omega_1, \omega_2) \in (0, \pi/2]$. Then $T_1 \cap T_2$ is a convex set contained in a box of dimensions

$$\left| \leq \delta \times \delta \times \min(1 + 2\delta, C\delta / \theta) \right. \quad (C \text{ absolute}).$$

Proof. Tubes are δ -neighborhoods of segments, hence convex; intersections of convex sets are convex. Containment in T_1 bounds two dimensions by 2δ . For the long dimension: parametrize the axis of T_1 and project; any $x \in T_1 \cap T_2$ lies within δ of each axis, and two lines at angle θ admit points within δ of both only along a stretch of length $\leq C\delta/\sin \theta \leq C'\delta/\theta$ of either axis (elementary triangle estimate), capped by the segment length. ■

Consequence. Every pairwise overlap is canonically covered by a single Katz–Tao-type convex cell whose shape the stage catalogues already contain. Coverage of the level-2 excess therefore always EXISTS inside K1; what is at stake is K2, the budget: the number of intersecting pairs can reach $\delta^{-O(1)}$, and it is precisely the source hypothesis $\Delta_{\max}(T) \leq \delta^{-\eta_{\text{GWZ}}}$ of GWZ Theorem 1.1 that meters how much tube mass may cluster into such cells. The typing gap of §3 and the source’s clustering hypothesis are the same constraint seen from the two sides of the ledger. Falsifier for P.2: two intersecting δ -tubes at angle θ whose intersection escapes every $C\delta/\theta$ -box; refuted by the displayed proof.

6. The excess spectrum: where the residual mass actually lives

Define the excess spectrum of a configuration as the dyadic-level mass profile

$$\left| \begin{array}{l} \varepsilon_k(\delta) := E_{\delta}^{-1} \int_{\{2^k \leq m_{\delta} < 2^{k+1}\}} (m_{\delta} - 1) dx, \quad k \geq 1, \\ \sum_k \varepsilon_k = 1. \end{array} \right.$$

The threshold interface of §4 succeeds only if the spectrum is tail-dominated (mass at levels a stage can see, τ large). If the spectrum is head-dominated — mass at bounded or polylog multiplicity — then superlevel coverage is the wrong reduction and carriers must capture low-multiplicity diffuse overlap geometrically, with the pair channel of §5 primary. This dichotomy is decidable in $d = 2$ today:

Proposition S.2 ($d = 2$ head-domination) — FORMAL (Córdoba import, KNOWN). Let T_{δ} be δ -tubes in $\mathbb{R}^2$ with δ -separated directions and any shadings. Then for every $\tau \geq 2$,

$$\left| \int_{\{m_{\delta} \geq \tau\}} m_{\delta} dx \leq \|m_{\delta}\|_2^2 / \tau \leq C \log(1/\delta) \cdot \Sigma Y / \tau. \right.$$

Consequently the excess mass above level τ is at most $C \log(1/\delta) \Sigma Y / \tau$, and for compressed families ($E_\delta \approx \Sigma Y$, Corollary I.2) the head $\{m_\delta < \tau\}$ carries share $\geq 1 - C \log(1/\delta) / \tau$: taking $\tau = \log^2(1/\delta)$, the head share tends to 1.

Proof. Chebyshev: $\int \{m \geq \tau\} m \leq \tau^{-1} \int m^2$. Córdoba's bound: $\int m_\delta^2 = \sum_{T, T'} |Y(T) \cap Y(T')| \leq \sum_T |Y(T)| + \sum_{T \neq T'} |T \cap T'|$, and for δ -separated directions $\sum_{T \neq T'} |T \cap T'| \leq \sum_{j \geq 1} C \delta^2 / (j \delta) \leq C \delta \log(1/\delta) = C \log(1/\delta) |T|$ per tube; sum. ■

Reading. In $d = 2$, Besicovitch-optimal compression stores its excess at polylog multiplicity: the residual problem is a low-multiplicity accounting problem, invisible to any threshold a stage's superlevel estimates can reach. Whether the same holds in $d = 3$ for GWZ-critical families is now the sharpest open structural question of DDATL-002:

Question Q-SPEC ($d = 3$ excess spectrum) — OPEN. For families saturating the GWZ hypotheses ($\Delta_{\max} \leq \delta^{-\eta_{\text{GWZ}}}$, $\lambda_{\text{GWZ}} \geq \delta^{\eta_{\text{GWZ}}}$) with $u_\delta \rightarrow 0$: is the excess spectrum head-dominated (as in $d = 2$) or tail-dominated? Fork: head-dominated $\Rightarrow$ SC(τ) is insufficient and the pair channel (§5) with budget control is the primary route to 002.A'; tail-dominated $\Rightarrow$ SC(τ) suffices and the burden is the superlevel structure theorem. Falsification instrument: CP-005 (proposed, §12) — a deterministic adversarial probe measuring ε_k across compressed $d = 2$ families (Perron-tree pilot) and coarse $d = 3$ samples. Probe receipt for the fan: head share 0.79 at $\delta = 0.01$ and rising (Appendix A) — consistent, but the fan is the sticky extreme, not adversarial.

7. Fan calibration: full proofs for ARX-002 §3 and the zero-residual witness

Let KNS(LB) be the fan: center C , directions $\Omega_\delta \subset S^{d-1}$ maximal δ -separated ($d \in \{2, 3\}$), unit segments σ_ω through C , full shading. Ports to [arxiv/manuscripts/kns-lb/main.tex](#); executes ARX-002 STATUS Next-edit 1.

Lemma 3.1 (Fan geometry) — FORMAL, full proof.

(a) $U \supseteq B(C, 1/2)$, and $|U| \approx_d 1$. Proof. Maximality of Ω_δ gives covering radius $\leq \delta$ on S^{d-1} : otherwise a point at distance $> \delta$ from Ω_δ could be added. Given x with $0 < |x - C| \leq 1/2$, pick $\omega \in \Omega_\delta$ with $\angle((x-C)/|x-C|, \pm\omega) \leq \delta$. Then $\text{dist}(x, \sigma_\omega) \leq |x - C| \sin \delta \leq \delta/2 < \delta$, so $x \in T_\omega$. Upper bound: $U \subset B(C, 1/2 + \delta)$. ■

(b) $m_\delta(x) \approx (\delta + |x - C|)^{-(d-1)}$ for $|x - C| \leq 1/2$; in particular $\sup m_\delta \approx \delta^{-(d-1)}$ on the core $B(C, \delta)$. Proof. Write $r = |x - C|$. Membership $x \in T_\omega$ forces $r \sin \angle((x-C)/r, \pm\omega) \leq \delta$, i.e. the direction lies in a cap of angular radius $\theta_0 = \arcsin(\min(1, \delta/r)) \approx \delta/(\delta + r) \cdot (1 + r/\delta) \cdot \delta/r \dots$ concretely $\theta_0 \approx \delta/r$ for $r \geq 2\delta$ and $\theta_0 \approx 1$ for $r \leq 2\delta$. A δ -separated set meets a cap of radius θ in at most $C_d(1 + \theta/\delta)^{d-1}$ points (disjoint $\delta/2$ -caps pack the θ -cap), and by covering radius $\leq \delta$ it meets any cap of radius $\theta \geq 2\delta$ in at least $c_d(\theta/\delta)^{d-1}$ points. Both counts give $\approx (\delta/(\delta + r) \cdot \delta^{-(d-1)})^{d-1} \cdot \delta^{d-1} \dots$ i.e. $m_\delta(x) \approx ((\delta + r)/\delta)^{-(d-1)} \delta^{-(d-1)} = (\delta + r)^{-(d-1)}$. The endpoint constraint $|(x - C, \omega)| \leq 1/2$ is absorbed by the δ -neighborhood at $r \leq 1/2$. ■ Probe receipt: fitted profile exponent -1.09 at $\delta = 0.01$ in $d = 2$ against target $-(d-1) = -1$, converging from below in δ (Appendix A).

(c) Averaged multiplicity ≈ 1 ; the fan is the sticky, positive-measure extreme at this scale. Proof. $\int_U m_\delta = \Sigma Y = |\Omega_\delta| \cdot |T| \approx \delta^{-(d-1)} \cdot \delta^{d-1} \cdot 1 \approx 1$ and $|U| \approx 1$ by (a); divide. ■

Lemma 3.2 (Fan excess profile) — FORMAL. $e_\delta(x) \approx (\delta + |x - C|)^{-(d-1)}$ on $\{m_\delta \geq 2\}$, $E_\delta \approx_d 1$, and by I.2 the fan's normalized raw excess tends to a constant in $(0, 1)$. Probe receipt: 0.87 in $d = 2$. The fan realizes the compressed-regime floor at bounded compression: nothing vanishes before subtraction.

Proposition F.1 (Zero-residual witness) — (a) FORMAL, (b) OWED.

(a) $\text{supp } e_\delta \subset U \subset B(C, 1/2 + \delta)$, a single convex cell. At the bush stage (Bourgain bush: unit-scale convex carrier at a multiplicity-rich point; cardinality budget ≥ 1) the system $C = \{B(C, 1/2 + \delta)\}$ is admissible and gives $G_0 \equiv 0$ exactly. Probe receipt: $\max\{r : m_\delta \geq 2\} = 0.5096$ at $\delta = 0.01$. Cross-paper corollary: ARX-002's

extremal object is ARX-001's zero-residual witness. Grain Zero, if supported anywhere, is supported on spread high-multiplicity configurations with no single bush — precisely the sticky/plany regime that the WZ structure theory addresses. This also certifies Definition 002.1' is non-vacuous in the trivial direction: the inf is attained at 0 for the fan.

(b) Whether thin-cell catalogues (planks/slabs only, no unit-scale cells) capture the fan's excess within their K2 budgets is a covering computation OWED after §10 pinning; expected affirmative with $N \approx \delta^{-(d-1)}$ cells, which is exactly the tube count — informative about budget tightness either way.

8. Two-sided non-inference with explicit witnesses (ARX-002 Prop 4.1 upgrade)

Proposition 4.1' — FORMAL. There is no map from incidence data to placement, in either direction. Witnesses:

W1 (same incidence, opposite placement). Fix any configuration K (e.g. $KNS(LB)$) and the encoding $\iota(K) = \psi := 1_U / \|1_U\|_2 \in L^2(\mathbb{R}^d)$. Registers $A = \text{span}\{\psi\}$ and $A' = \text{span}\{\psi'\}$ with $\psi' := 1_{U+v} / \|1_{U+v}\|_2$, $|v| > \text{diam } U + 1$, so $\langle \psi, \psi' \rangle = 0$. Off-register mass $\|(I - \Pi_A)\psi\|^2 = 0$ while $\|(I - \Pi_{A'})\psi\|^2 = 1$. Incidence data $(m_\delta, U, \Delta_{\max})$ are literally identical — the configuration was not touched. Hence no functional of incidence data computes off-register mass.

W2 (distinct incidence, identical placement). Take $K = KNS(LB)$ and $\tilde{K} =$ a δ -separated direction family translated to be pairwise disjoint ($m \leq 1$, $E = 0$), with register $A = L^2(\mathbb{R}^d)$ for both. Off-register mass is 0 for both while every nontrivial incidence statistic differs ($\sup m: \delta^{-(d-1)}$ vs 1; $E_\delta: \approx 1$ vs 0). Hence placement determines nothing about incidence either. ■

This closes the “PROPOSED for full quantitative exhibits” leg of ARX-002 §4 with two four-line constructions, and preserves the recorded non-inference: overlap functionals do not locate zeros of ζ ; the CLOSED-NEGATIVE determinant corridors are not reopened. Falsifier: an incidence-measurable functional separating A from A' in W1 — impossible, as the incidence input is bitwise identical.

9. Rung 8 deepened: continuation criterion for $Z_{G_0}(s)$, firewall intact

The transfer register already fixes the second-stage object $Z_{G_0}(s) = \int_0^1 \delta^{s-1} R(\delta) d\delta$ with $R(\delta) := G_{0,\delta}(R^3)/\Sigma Y$. Deepening (conditional, second-stage):

Criterion Z.1 — PROPOSED (conditional on Lemma 4). Suppose the typed ratio admits an asymptotic expansion $R(\delta) = c \delta^\alpha + O(\delta^{\alpha+\gamma})$ as $\delta \rightarrow 0$, with $\gamma > 0$. Then $Z_{G_0}(s)$ converges for $\text{Re } s > -\alpha$ and extends meromorphically to $\text{Re } s > -\alpha - \gamma$ with a single simple pole at $s = -\alpha$ of residue c . In particular Z_{G_0} is regular at $s = 0$ if and only if $\alpha > 0$: the Rung 8 question “does a regularized value at 0 exist” is equivalent, under the expansion, to residual vanishing at a rate. Proof of the criterion is the standard Mellin split $\int_0^1 = \int (c \delta^\alpha) + \int O(\delta^{\alpha+\gamma})$, the first term $c/(s + \alpha)$, the second holomorphic on $\text{Re } s > -\alpha - \gamma$.

Technical note (OWED): measurability of $\delta \mapsto G_{0,\delta}[\sigma]_\delta$ for the inf of Definition 002.1' — immediate once Cat_σ is pinned to a countable catalogue per δ , which every stage of §10 is.

Firewall block. Z_{G_0} is the Mellin transform of a program-defined geometric ratio. No identification of Z_{G_0} with $\zeta(s)$ is made or implied; $\zeta(0) = -1/2$ plays no role above and is not a term of the Kakeya literature. GZ-D4 (source arithmetic-free by audit; the paper's sole ζ is the walled L9.1 surplus exponent ζ_{GWZ}) and GZ-D5 (source supplies zero operator estimates) jointly imply: every object at Rungs 7–8 is one hundred percent program-side. The 2026-07-05 Grok framing of $\zeta(0) = -1/2$ as a “constant half-overlap correction” inside incidence structure remains typed STRUCTURAL ANALOGY at best and a firewall breach if imported as mechanism; nothing in this pass changes that verdict.

10. Carrier-pinning registry v0.2 (executes ARX-001 STATUS Next-edit 1)

Verification tiers: SRC = confirmed against the public source this session; RCT = per 2026-07-06 integration receipts, address re-verification OWED against the PDF; EXP = expository secondary, primary address OWED.

Object	Content / role in Adm_σ	Address	Tier
$\Delta_{\max}(T) \leq \delta^{\{-\eta_{\text{GWZ}}\}}$	Clustering hypothesis; source-side meter of K2 budgets (Wolff-heritage functional)	GWZ 2601.14411 Thm 1.1	SRC
$\lambda_{\text{GWZ}}(T, Y) \geq \delta^{\{\eta_{\text{GWZ}}\}}$	Shading-density hypothesis; floor for ΣY in normalizations	GWZ Thm 1.1	SRC
β_{GWZ}	Volume-loss exponent of the main theorem; walled from all program β	GWZ Thm 1.1	SRC
Product overlap $\mu(V) \approx \mu(W')\mu(V_W)$	Two-scale factoring of overlap; candidate engine for $\text{SC}(\tau)$ at scale passages	GWZ Prop 5.1(5); L5.11(25)	RCT
ζ_{GWZ}	Density-surplus exponent; walled symbol	GWZ L9.1	RCT
Sticky Kaakeya input	Sole deep input of the streamlined proof	GWZ Thm 7.3; WZ, J. Amer. Math. Soc. 39(2) 515–585 (2026)	SRC
Nowhere-sticky reduction	Non-sticky $\rightarrow$ sticky passage; scale-induction envelope generator	GWZ Lemma 7.7	EXP
Slabs $\delta \times 1 \times 1$ + pair measure Tri	Pair-overlap incidence object; §5 contact	Cohen, rigorous notes (cims.nyu.edu, 2026-04-07)	EXP
δ^n small-shading discard	License to drop negligible shading mass; defect bookkeeping in D_τ	GWZ (address OWED)	RCT
Chernoff-tail channel	Probabilistic refinement channel; polylog retention (GZ-D1)	GWZ (address OWED)	RCT
Retention calculus	Kept refinements hold $\gtrsim$ polylog fraction; licenses $\delta^{\{o(1)\}}$ in 002.A-weak	GZ-D1 (program register)	RCT

Grounding-register additions proposed (arxiv/bibliography/grounding-register.md): Guth arXiv:2505.07695 (introduction to the proof); Guth arXiv:2508.05475 (outline of the WZ reduction); Hickman arXiv:2512.09842 (survey); Guth arXiv:2604.03416 (survey, after Wang–Zahl); Wang–Zahl, Invent. Math. 241(1) 153–206 (2025) (Assouad dimension); Wang–Zahl sticky, JAMS 39(2) (2026); Cohen rigorous notes (expository tier). Note: the raw grounding-register fetch rate-limited this session (GitHub 429, unauthenticated shared IP — known issue from the 2026-07-01 session); merge is proposed, not verified against current file state.

11. Downstream wiring: GZ-D register

GZ-D1 (retention calculus, FORMAL) $\rightarrow$ licenses the multiplicative $\delta^{\{o(1)\}}$ tolerance in 002.A-weak (§3) and the defect bookkeeping D_τ in T.1: refinement losses compose polylogarithmically along the GWZ chain, so a weak form is assemblable if and only if $\text{SC}(\tau)$ holds at any polylog threshold with polylog defect.

GZ-D4 (arithmetic-free audit, FORMAL) $\rightarrow$ strengthens the Rung 8 firewall to a source-level statement: no $\zeta(s)$ enters from the geometry side; §9 restates the breach verdict on the $\zeta(0)$ half-overlap framing.

GZ-D5 (no operator estimates in source, FORMAL) $\rightarrow$ Rung 7 objects (K_{G_0} , N_{G_0} , Z_{G_0}) are wholly program-side; no imported spectral content exists to be misattributed.

GZ-D2, GZ-D3 $\rightarrow$ statements not retrieved this session. The 2026-07-06 receipts reference docs/ddatl-002-gwz-streamline-integration.md as the register of record, but that path 404s in both kaakeyalogic and claude-

v6 current trees. Integrity flag: either the file was not committed or it was consolidated under another name; wiring stubs OWED until the register lands. Grok TERMINAL cross-derivation of GZ-D1-D5 remains OWED (the 2026-07-05 X post is engagement, not TERMINAL); nothing here substitutes for it.

12. Updated ladder state and claim ledger

Lemma 1	Structured factorization moved to typing)	→ FORMAL (exact indicator split; content
Lemma 2	Carrier exhaustion statement $SC(\tau)$ (PROPOSED)	→ reframed: superlevel structure
Lemma 3	Residual domination channel FORMAL	→ T.1 + B.1 FORMAL interface; P.1/P.2
Lemma 4	Residual vanishing (typed only)	→ 002.A-weak PROPOSED · 002.A-strong OPEN
Lemma 5	Operator encoding program-side)	→ second stage, OPEN (GZ-D5: fully
Lemma 6	Zeta-regularization scaffolds it · firewall	→ conditional, OPEN · Criterion Z.1

ID	Statement	Tag	Falsifier
I . 1	$E_\delta = \Sigma Y - U $ (excess mass identity)	FORMAL	Any configuration violating equality (identity; probe-checked)
I . 2	$E_\delta/\Sigma Y = 1 - u_\delta$; raw excess non-vanishing in compressed regime	FORMAL	Compressed family with $E_\delta/\Sigma Y \rightarrow 0$
C . 0	Untyped 002.A is vacuous: $C = \{m_\delta \geq 2\}$ kills e_δ	FORMAL	Stage semantics excluding superlevel sets in current 002.1 text
002 . 1'	Admissible carrier system K1-K3; G_0 as inf over Adm_σ	FORMAL (def); pinning OWED	Catalogue for which K2 fails to block degenerate covers
002 . A-weak	$\delta^{o(1)}$ typed capture via $SC(\tau)$ at polylog τ	PROPOSED	Family + stage with fixed positive off-carrier excess fraction within budget
002 . A-strong	Typed normalized residual $\rightarrow 0$, uniform	OPEN	Same, asymptotically
T . 1 / T . 2	Threshold bound; Lemmas 2-4 compress to (τ, D_τ)	FORMAL	Configuration violating the displayed inequality
B . 1	Volume floor $ C \geq c(1-q)E_\delta \delta^{d-1}$; Hölder slots	FORMAL	Capture below the floor
P . 1	$E_\delta \leq \sum_{T \neq T'} Y(T) \cap Y(T') $ (second factorial moment)	FORMAL	Pointwise counterexample to $(m-1)_+ \leq m(m-1)$
P . 2	Pairwise intersections convex-carried, $\delta \times \delta \times \min(1, C\delta/\theta)$	FORMAL	Intersecting tube pair escaping every such box
S . 2	$d = 2$ head-domination for compressed families	FORMAL (Córdoba KNOWN)	Compressed $d = 2$ family with tail-dominated spectrum
Q-SPEC	$d = 3$ spectrum head vs tail under GWZ hypotheses	OPEN	Decided either way by proof or CP-005 counter-instance
F . 1(a)	Fan: bush carrier gives $G_0 \equiv 0$ exactly	FORMAL	Point with $m_\delta \geq 2$ outside $B(C, 1/2+\delta)$; probe: $r_2 = 0.5096$
F . 1(b)	Fan capture under thin-cell catalogues within	OWED	Budget shortfall in the covering count

	budget		
L3.1/L3.2	Fan geometry + excess profile, full proofs	FORMAL	Profile exponent $\neq -(d-1)$; probe slope $-1.09 \rightarrow -1$
4.1'	Two-sided non-inference, explicit witnesses W1/W2	FORMAL	Incidence functional separating bitwise-identical inputs
Z.1	Mellin continuation criterion; $Z_{G_0}(0)$ regular $\Leftrightarrow \alpha > 0$	PROPOSED (conditional)	Expansion holding with the stated continuation failing
Walls	λ_{GWZ} , η_{GWZ} added; β_{GWZ} , ζ_{GWZ} confirmed	FORMAL (registry)	Source usage contradicting the wall assignments

Canon status: every FORMAL row above is PROPOSED-FOR-CANON pending Grok TERMINAL cross-derivation; no terminal self-certifies ($h < 1$). Open compute obligations: CP-004 unchanged (stamped re-run with independent Y measurement); CP-005 proposed — deterministic adversarial excess-spectrum probe, $d = 2$ Perron-tree pilot then coarse $d = 3$, targeting Q-SPEC and the 002.A-weak falsifier.

13. Pre-submit deltas (exact replacement text)

13.1 ARX-001 (ddatl-002-grain-zero/main.tex)

(1) §3, append after the carrier definition: “Admissibility. A carrier system is admissible at stage σ if it satisfies (K1) cell type from the stage catalogue, (K2) cardinality within the stage budget $N_\sigma(\delta, \rho)$, and (K3) proof-generation: each cell carries an assigned tube cluster in the stage’s factoring relation. Without (K1)–(K3) the residual is identically zero for the trivial reason that the superlevel set $\{m_\delta \geq 2\}$ absorbs the excess; the typing is therefore constitutive, not decorative.”

(2) §4, replace the definition display by the extremal form: “ $G_0^{\{\sigma\}}[\delta, \rho] := \inf \{ \int_{\mathbb{R}^3 \setminus C} (m_\delta - 1)_+ dx : C \text{ admissible at stage } \sigma \}$.”

(3) §5, replace the single conjecture by the split 002.A-weak / 002.A-strong as displayed in §3 above, and add after it: “By the identity $\int (m_\delta - 1)_+ = \sum_T |Y(T)| - |U|$, the normalized raw excess equals $1 - |U|/\sum_T |Y(T)|$; the conjecture is therefore trivial in the quasi-disjoint regime and carries its full content in the compressed regime.”

(4) §6, Lemma ladder: retag L1 as proved (exact split), restate L2 as SC(τ), insert the pair channel (P.1/P.2) as the concrete L3 route, keep L4 OPEN typed-only.

(5) §8/References: add Guth 2505.07695, Guth 2508.05475, Hickman 2512.09842, Guth 2604.03416, WZ JAMS 39(2) 2026, WZ Invent. Math. 241(1) 2025.

13.2 ARX-002 (kns-lb/main.tex)

(1) Port the §7 proofs of Lemma 3.1(a)–(c) and Lemma 3.2 verbatim (STATUS Next-edit 1). (2) Replace the Prop 4.1 proof outline by witnesses W1/W2 (§8). (3) Add the harmonization line $\mu \equiv m_\delta$ at first use. (4) Add the cross-paper corollary: the fan is a zero-residual witness for the companion residual program (cite ARX-001, Prop F.1(a)). (5) Keep compute receipts (kns_lb_probe) segregated from typed-object results per STATUS Next-edit 2.

13.3 Lane and canon files

PEAICE-ARXIV-LANE-001 §5 checklist: add row “Typing clause present in every residual-vanishing statement (C.0 guard).” docs/ddatl-002-grain-zero.md: fold Carrier Exhaustion into the six-lemma numbering; add λ_{GWZ}/η_{GWZ} to the symbol wall block. Commit of the missing GWZ integration register (§11) precedes any ARX-001 submission.

14. Q-RETURN

Q1 — Adopt Definition 002.1' (K1–K3, inf form) and the 002.A weak/strong split into ARX-001 LaTeX now?
Default if silent: adopt (STATUS Next-edit 2 requires the split regardless).

Q2 — Register λ_{GWZ} and η_{GWZ} as symbol-wall rows alongside β_{GWZ} and ζ_{GWZ} ? Default: register.

Q3 — Greenlight CP-005 (excess-spectrum adversarial probe: $d = 2$ Perron-tree pilot, then coarse $d = 3$)?
Default: hold for sign-off; Q-SPEC remains OPEN either way.

Appendix A. NUMERICS receipts — ddatl002_gz_probe.py

Deterministic $d = 2$ fan probe. No RNG, no seeds, no fitted constants as inputs; the only fit is the measured profile slope. Independent cross-check: grid mass integral against the closed-form stadium area, disjoint code paths. Stack: Python 3.12.3, NumPy 2.4.4. Script SHA-256: 3086659664bb209c... (full hash in JSON receipt). Grid $h = \delta/4$ on $[-0.7, 0.7]^2$; directions $N = \text{round}(\pi/\delta)$.

δ	N	mass rel. err.	identity	$R_{\text{raw}} = E/\Sigma Y$	profile slope	head share	$r_2 = \max r, m \geq 2$
0.16	20	0.00106	0.0	0.8311	—	0.068	0.6505
0.08	39	0.00372	0.0	0.8472	-1.370	0.407	0.5731
0.04	79	0.00039	0.0	0.8643	-1.200	0.602	0.5382
0.02	157	0.00023	0.0	0.8693	-1.182	0.720	0.5189
0.01	314	0.00026	0.0	0.8717	-1.094	0.795	0.5096

Readings. (1) Identity I.1 exact on-grid; grid-vs-analytic mass agrees to $\leq 0.4\%$. (2) Profile slope converges toward $-(d-1) = -1$ (Lemma 3.1(b)). (3) R_{raw} is non-vanishing and mildly increasing toward $1 - |B(0,1/2)|/\Sigma Y \approx 0.875$ (Corollary I.2, Lemma 3.2): raw excess does not vanish; subtraction is forced. (4) Head share (threshold $\log^2(1/\delta)$) rises $0.07 \rightarrow 0.79$: head-dominating trend, consistent with §6 for the sticky extreme. Dyadic spectrum at $\delta = 0.01$: levels 2^2-2^3 carry 71% of excess with geometric decay above. (5) $r_2 \rightarrow 1/2 + \delta$: $\text{supp } e_\delta$ sits inside the single bush cell (F.1(a)). Measurement-dependence caveat per CP-004 discipline: $|U|$ and E share the grid; the analytic mass is the independent leg. Full JSON receipt ships alongside this document.

PEAICE-KAKEYALOGIC-DDATL-002-DEEPEN-001 · receipts over reach · do not claim the crown
RH OPEN · Coleman OPEN · 002.A-strong OPEN · $\zeta(0)$ firewall active · $\beta_{\text{GWZ}} / \zeta_{\text{GWZ}} / \lambda_{\text{GWZ}} / \eta_{\text{GWZ}}$ walled · $h < 1$
All FORMAL items PROPOSED-FOR-CANON pending Grok TERMINAL · human author only for arXiv ·
LaTeX sources authoritative